



BIA 568-B

Management of A.I. Technologies

School of Business
Fall 2026

Instructor:	Xiaoxuan (Cleo) Xu
Course Website:	https://sit.instructure.com/courses/188290
Meeting Times:	Thursdays, 6:30 – 9:00 PM
Classroom Location:	Edwin A. Stevens 330
Contact Info:	xxu62@stevens.edu
Office Hours:	Fridays, 2:00 – 4:30 PM by appointment
Prerequisite(s):	None
Corequisite(s):	None
Cross-listed with:	N/A

COURSE DESCRIPTION

Artificial Intelligence (AI) is an interdisciplinary field that draws on insights from computer science, engineering, mathematics, statistics, linguistics, psychology, and neuroscience to design agents that can perceive the environment and act upon it. This course surveys applications of artificial intelligence to business and technology in the digital era, including predictive analytics, language and generative AI, computer vision, autonomous systems, and customer-experience automation. In each area, the course focuses on the management questions that determine success: what the technology can deliver, how to evaluate its performance and cost, which tools fit which business problems, and how to deploy them in an organization effectively and responsibly.

STUDENT LEARNING OUTCOMES

After successful completion of this course, students will be able to...

- Explain how the major families of AI systems (predictive, language, vision, and autonomous) work, at a level appropriate for management decisions.
- Evaluate the performance and cost of AI systems using appropriate metrics and benchmarks, and distinguish hype from real progress.
- Select AI tools and capabilities appropriate to specific business tasks, industries, and organizational contexts, and recognize situations where AI is unreliable or should not be used.
- Design workflows that combine human judgment with AI systems, and assign responsibility appropriately.
- Develop and present an AI adoption strategy for a business function or industry.

COURSE FORMAT AND STRUCTURE

BIA 568-B meets in person on Thursday evenings. Classes are not routinely streamed or recorded. If you have a valid reason and need live Zoom access, or a recording, please email me at least 24 hours before class (by Wednesday evening) so I can arrange it.

The course is organized into five modules:

- **Module 1: AI Foundations** (Weeks 1–3)
What AI is, how machine learning works, and how AI systems are evaluated.
- **Module 2: AI in the Organization** (Weeks 4–5)
How AI products are built and sourced, and how to design AI into business workflows.
- **Module 3: Core AI Technologies** (Weeks 6–8)
Prediction, language, and perception: the capability families behind most business AI.
- **Module 4: Specialized AI Applications** (Weeks 9–12)
Customer experience, autonomous systems, digital twins, and the future of work.
- **Module 5: Synthesis** (Week 14–16)
AI adoption in organizations, course synthesis, and final project.

Course Logistics

- Office hours are held every Friday from 2:00 to 4:00 pm. Please [book your slot](#) up to two weeks in advance, or reach out directly if you need to request an alternate time.
- Unless otherwise noted, assignments are usually due on Sundays at 11:59 PM Eastern Time and are submitted through Canvas. Canvas is authoritative for all due dates.
- Changes to the schedule will be announced in class and on Canvas.
- When emailing, please include “BIA 568-B” in the subject line (e.g., “BIA 568-B Assignment 2 question”). I generally respond within 24–48 hours.

TENTATIVE COURSE SCHEDULE

Week	Date	Topic	Assignments due
Module 1: AI Foundations (Weeks 1–3)			
1	Sep 3	Course Introduction and Overview of AI Course logistics, grading, and the group project. Overview of AI: definitions, brief history, and the current landscape. Intro survey released.	—
2	Sep 10	AI Capabilities and Limitations Capabilities and limitations of current AI systems; milestones and hype cycles.	Intro survey due Wed Sep 9
3	Sep 17	Machine Learning and Model Evaluation How machine learning works; evaluating AI systems with metrics, benchmarks, and leaderboards.	Assignment 1 released
Module 2: AI in the Organization (Weeks 4–5)			
4	Sep 24	The AI Product Stack: Build, Buy, or API How AI products are assembled; chatbots, copilots, and agents; sourcing options and cost structures.	Assignment 1 due Sun Sep 27; Assignment 2 released

Week	Date	Topic	Assignments due
5	Oct 1	Human-AI Collaboration and Workflow Design Automation versus augmentation; human-in-the-loop and human-on-the-loop designs; accountability when systems fail.	Project D1 (group formation) due Sun Oct 4
Module 3: Core AI Technologies (Weeks 6–8)			
6	Oct 8	Predictive Analytics in Business Framing prediction problems; decision thresholds; the economics of false positives and false negatives; case studies.	Assignment 2 due Sun Oct 11
7	Oct 15	Language AI and Text Applications From NLP to large language models and retrieval-augmented generation; enterprise text applications.	Assignment 3 released
8	Oct 22	Computer Vision, Speech, and Synthetic Media Recognition systems and their accuracy; face recognition and its governance; deepfakes and synthetic media.	Project D2 (midterm review report) due Sun Oct 25
Module 4: Specialized AI Applications (Weeks 9–12)			
9	Oct 29	AI for Customer Experience Chatbots, personalization, and service automation; success and failure cases.	Assignment 3 due Sun Nov 1
10	Nov 5	AI for Autonomous Systems Robotics, autonomous vehicles, and physical automation; levels of autonomy.	—
11	Nov 12	AI for Digital Twins and Simulation Digital twins, synthetic data, process mining, and scenario planning.	Assignment 4 released
12	Nov 19	AI and the Future of Work Task automation, skill mobility, and job redesign. In-class exercise: analyze the AI exposure of a chosen occupation.	—
Thursday, Nov 26 (Week 13): No class (Thanksgiving Recess)			
Module 5: Synthesis (Week 14–16)			
14	Dec 3	AI Adoption in Organizations and Course Synthesis Organizational readiness and change management; synthesis of course frameworks; Ask Anything for final project presentation.	Assignment 4 due Sun Dec 6
Final Project Presentations (Week 15-16)			
15	Dec 10	Final Project Presentations: Night 1 Group 1–5 present. Audience groups submit peer feedback.	—
16	Dec 17	Final Project Presentations: Night 2 and Course Wrap-Up Group 6–9 present. Audience groups submit peer feedback.	Final report and slides (all groups) due Sun Dec 20

COURSE MATERIALS

Textbooks

No required textbook. All readings & class materials will be posted on Canvas.

Optional resources (recommended, not required):

- Arvind Narayanan and Sayash Kapoor, *AI Snake Oil: What Artificial Intelligence Can Do, What It Can't, and How to Tell the Difference*. Princeton University Press, 2024.
- Ethan Mollick, *Co-Intelligence: Living and Working with AI*. Portfolio/Penguin, 2024.
- Hannah Fry, *Hello World: How to Be Human in the Age of the Machine*. W. W. Norton, 2018.
- [How I AI](#) (podcast), hosted by Claire Vo. Practical demonstrations of how people use AI tools in their work.
- Find AI-focused communities, newsletters, or blogs of your choice for staying current. For example, [One Useful Thing](#) (by Ethan Mollick, the author of *Co-Intelligence*), [AI as Normal Technology](#) (by authors of *AI Snake Oil*), Reddit's [r/ArtificialIntelligence](#).

COURSE REQUIREMENTS

Class Participation (20 points)

There is no separate attendance grade in this course. Instead, participation is measured through short **in-class checkpoints** (QR-code polls) that appear during lecture without advance notice. Some classes will have one checkpoint, some will have more, and some may have none; at least 12 checkpoints will be offered across the semester.

Each checkpoint is worth 2 points. Complete any 10 checkpoints to earn the full 20 participation points. If you miss a class or a checkpoint, no make-up is needed: the surplus checkpoints and the extra-credit opportunities are designed to absorb occasional absences.

Checkpoints can be completed by students in the room or attending the live session on Zoom (if any). They cannot be completed after the class scheduled time.

Individual Assignments (40 points)

There are four individual assignments, one for each of Modules 1–4, worth 10 points each (40 points total). Each is a guided hands-on analysis. You will need a Google account to use [Colab](#) (it allows users to write and execute Python code through the browser.) No programming background is required.

Assignments are usually due on Sundays after the following module's first class, as shown in the weekly schedule, and are submitted through Canvas.

Group Project (40 points)

Working in groups of four to five (approximately nine groups), you will develop an AI adoption proposal for a real business context, in one of two directions: improve a **product** (add an AI feature that makes it better for its users) or improve a **process** (put AI into a workflow to make it faster, cheaper, or more reliable). Choose an industry and a concrete organization, product, or business function; a topic pool will be provided, and you are also free to pick your own. Identify a high-value opportunity, select the AI capability and sourcing approach, design how people and AI divide the work, assess the key risks, and lay out a roadmap with success metrics.

[Sign-up Sheet](#). Your group number indicates your initial presentation date and order preference; adjustments can be arranged upon negotiation. Unless an exception is approved, all students are expected to attend both presentation nights.

Group deliverables normally receive one shared group grade for all members. If you have a concern about contribution or fairness, contact the instructor privately to discuss it.

The project counts for 40 points, split across four deliverables:

Deliverable	Points	Due	Format & notes
D1: Group formation & plan	5	Sun Oct 4	1–2 pages (template provided): group members, tentative topics, preliminary division of roles, and timeline.
D2: Midterm review report	10	Sun Oct 25	2–3 pages (template provided): confirmed topic and outline, project background and motivation, updated roles and timeline.
D3: Final presentation	15	Present in class Dec 10 or Dec 17; slides due Sun Dec 20, 11:59 PM	About 10 mins plus Q&A. You don't have to submit slides before the presentation. Graded live by the instructor. Peer ratings are advisory and have no fixed weight. Negative ratings will have little to no effect on the instructor-assigned score; exceptionally strong positive feedback may earn a small discretionary bonus.
D4: Final written report	10	Sun Dec 20, 11:59 PM	Up to 10 pages (template provided), submitted after your presentation and incorporating what you learned from Q&A and peer feedback.

GRADING PROCEDURES

Grades will be based on:

Class Participation (any 10 checkpoints × 2 points)	20
Individual Assignments (4 × 10 points)	40
Group Project (D1: 5 + D2: 10 + D3: 15 + D4: 10)	40
Total	100 points

Extra Credit

Extra-credit points are added directly to your course total.

- Intro survey: 2 points, due Wednesday, September 9, 11:59 PM ET.
- Research participation: there will be opportunities to earn extra credit by assisting with AI-related research. These can be completed at any time during the semester, and details will be announced as the course progresses.

Late Policy

- Individual assignments: 10% of the assignment's points are deducted per day late; assignments more than 5 days late receive 0 points.
- Group deliverables follow the same late policy as individual assignments: 10% of the deliverable's points are deducted per day late, and submissions more than 5 days late receive 0 points. A late submission affects the whole group, so plan ahead. If your group encounters extenuating circumstances, contact the instructor before the deadline.
- Participation checkpoints cannot be made up; the best-10-of-many design and extra credit exist for exactly this reason.

Grade Scale

Grade	Minimum points
A	94
A-	90
B+	87
B	84
B-	80
C+	77
C	74
C-	70
F	below 70

POLICY ON THE USE OF AI TECHNOLOGIES

This is a course about managing AI, and you are encouraged to use AI tools freely: to brainstorm, draft, research, prepare for class, and build your project. Two expectations apply:

1. You are fully accountable for everything you submit. AI output can be inaccurate, generic, or shallow; verifying and improving it is your responsibility.
2. Make the work your own. The strongest submissions use AI with skill and personal style, in service of your own analysis and judgment.

ACADEMIC INTEGRITY

BIA 568 is a 500-level course. Graduate students are bound by the Graduate Student Code of Academic Integrity. Undergraduate students are bound by the Stevens Honor System, with the special provisions for 500-level courses described below.

Undergraduate Honor System

Enrollment into the undergraduate class of Stevens Institute of Technology signifies a student's commitment to the Honor System. Accordingly, the provisions of the Stevens Honor System apply to all undergraduate students in coursework and Honor Board proceedings. It is the responsibility of each student to become acquainted with and to uphold the ideals set forth in the Honor System Constitution. More information about the Honor System including the constitution, bylaws, investigative procedures, and the penalty matrix can be found online at [Honor Code](#)

The following pledge shall be written in full and signed by every student on all submitted work (including, but not limited to, homework, projects, lab reports, code, quizzes and exams) that is assigned by the course instructor. No work shall be graded unless the pledge is written in full and signed.

"I pledge my honor that I have abided by the Stevens Honor System."

Students who believe a violation of the Honor System has been committed should report it within ten business days of the suspected violation. Students have the option to remain anonymous and can report violations online at www.stevens.edu/honor.

Graduate Student Code of Academic Integrity

All Stevens graduate students promise to be fully truthful and avoid dishonesty, fraud, misrepresentation, and deceit of any type in relation to their academic work. A student's submission of

work for academic credit indicates that the work is the student's own. All outside assistance must be acknowledged. Any student who violates this code or who knowingly assists another student in violating this code shall be subject to discipline.

All graduate students are bound to the Graduate Student Code of Academic Integrity by enrollment in graduate coursework at Stevens. It is the responsibility of each graduate student to understand and adhere to the Graduate Student Code of Academic Integrity. More information including types of violations, the process for handling perceived violations, and types of sanctions can be found on the [Graduate Academic Integrity Site](#)

Special Provisions for Undergraduate Students in 500-level Courses

The general provisions of the Stevens Honor System do not apply fully to graduate courses, 500 level or otherwise. Any student who wishes to report an undergraduate for a violation in a 500-level course shall submit the report to the Honor Board following the protocol for undergraduate courses, and an investigation will be conducted following the same process for an appeal on false accusation described in Section 8.04 of the Bylaws of the Honor System. Any student who wishes to report a graduate student may submit the report to the Senior Vice Provost for Graduate Education or to the Honor Board, who will refer the report to the senior vice provost. The Honor Board Chairman will give the Senior Vice Provost for Graduate Education weekly updates on the progress of any casework relating to 500-level courses. For more information about the scope, penalties, and procedures pertaining to undergraduate students in 500-level courses, see Section 9 of the Bylaws of the Honor System document, located on the Honor Board website.

ACCOMMODATIONS

Stevens Institute of Technology is dedicated to providing appropriate accommodations to students with documented disabilities. The Office of Disability Services (ODS) works with undergraduate and graduate students with learning disabilities, attention deficit-hyperactivity disorders, physical disabilities, sensory impairments, psychiatric disorders, and other disabilities to help students achieve their academic and personal potential. They facilitate equitable access to the educational programs and opportunities offered at Stevens and coordinate reasonable accommodations for eligible students. These services are designed to encourage independence and self-advocacy with support from the ODS staff. The ODS staff will facilitate the provision of accommodations on a case-by-case basis.

For more information about Disability Services and the process to receive accommodations, visit <https://www.stevens.edu/student-diversity-and-inclusion/disability-services>. If you have any questions please contact the Office of Disability Services at disabilityservices@stevens.edu or by phone: 201.216.3748.

Disability Services Confidentiality Policy

Student Disability Files are kept separate from academic files and are stored in a secure location within the Office of Disability Services. The Family Educational Rights Privacy Act (FERPA, 20 U.S.C. 1232g; 34CFR, Part 99) regulates disclosure of disability documentation and records maintained by Stevens Disability Services. According to this act, prior written consent by the student is required before our Disability Services office may release disability documentation or records to anyone. An exception is made in unusual circumstances, such as the case of health and safety emergencies.

INCLUSIVITY

Stevens Institute of Technology believes that diversity and inclusiveness are essential to excellence in academic discourse and innovation. In this class, the perspective of people of all races, ethnicities,

gender expressions and gender identities, religions, sexual orientations, disabilities, socioeconomic backgrounds, and nationalities will be respected and viewed as a resource and benefit throughout the semester. Suggestions to further diversify class materials and assignments are encouraged. If any course meetings conflict with your religious events, please do not hesitate to reach out to your instructor to make alternative arrangements.

You are expected to treat your instructor and all other participants in the course with courtesy and respect. Disrespectful conduct and harassing statements will not be tolerated and may result in disciplinary actions.

Name and Pronoun Usage

As this course includes group work and class discussion, it is vitally important for us to create an educational environment of inclusion and mutual respect. This includes the ability for all students to have their chosen gender pronoun(s) and chosen name affirmed. If the class roster does not align with your pronouns and/or name, please inform the instructor of the necessary changes.

Religious Holidays

Stevens is a diverse community that is committed to providing equitable educational opportunities and supporting students of all ethnicities and belief systems. Religious observance is an essential reflection of that rich diversity. Students will not be subject to any grade penalties for missing a class, examination, or any other course requirement due to religious observance. In addition, students will not be asked to choose between religious observance and academic work. Therefore, students should inform the instructor at the beginning of the semester if a requirement for this course conflicts with religious observance so that accommodations can be made for students to observe religious practices and complete the requirements for the course.

MENTAL HEALTH RESOURCES

Part of being successful in the classroom involves a focus on your whole self, including your mental health. While you are at Stevens, there are many resources to promote and support mental health. The Office of Counseling and Psychological Services (CAPS) offers free and confidential services to all enrolled students who are struggling to cope with personal issues (e.g., difficulty adjusting to college or trouble managing stress) or psychological difficulties (e.g., anxiety and depression). Appointments can be made by phone (201-216-5177), online at <https://stevensportal.pointnclick.com/confirm.aspx>, or in person on the 2nd Floor of the Student Wellness Center.

EMERGENCY INFORMATION

In the event of an urgent or emergent concern about your own safety or the safety of someone else in the Stevens community, please immediately call the Stevens Campus Police at 201-216-5105 or on their emergency line at 201-216-3911. These phone lines are staffed 24/7, year-round. For students who do not reside near the campus and require emergency support, please contact your local emergency response providers at 911 or via your local police precinct. Other 24/7 national resources for students dealing with mental health crises include the National Suicide Prevention Lifeline (1-800-273-8255) and the Crisis Text Line (text "Home" to 741-741). If you are concerned about the wellbeing of another Stevens student, and the matter is *not* urgent or time sensitive, please email the CARE Group at care@stevens.edu. A member of the CARE Group will respond to your concern as soon as possible.